



Physics-informed Kolmogorov-Arnold networks with residual-based adaptive distribution for scattered acoustic field prediction

Yi Ren¹ · Ligang Wang¹ · Haitao Ma¹

Received: 8 May 2025 / Accepted: 23 February 2026

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2026

Abstract

Physics-informed neural networks (PINNs) have proven to be a powerful tool for scattering simulation, leveraging physics principles to inform the learning process. When solving the acoustic scattering problem induced by the scatterer, conventional PINNs that rely on multilayer perceptrons (MLP) neglect the inherent acoustic field dependencies, thus failing to propagate globally the scattering characteristics and accurately capture nonlinear features near the scatterer boundary. Researchers often define explicit architectures or increase training parameter counts to capture these dependencies, yet this inevitably introduces either structural constraints or heightened computational costs. In this paper, we propose a novel framework based on the Kolmogorov–Arnold Network (KAN), termed PIKAN, to address this limitation. Leveraging KAN-driven plane wave expansion and the Sine activation function’s efficient plane wave learning capacity, PIKAN effectively captures both global dependencies and fine-grained scattering features in the scattered acoustic field. We evaluate the effectiveness and generalization performance of the PIKAN model across three typical scattering scenarios: high-frequency case, irregular scatterer, and multiple scattering, to validate its practicality. Experiments demonstrate that PIKAN achieves a superior balance between both accuracy and computational efficiency, when compared against state-of-the-art baselines and alternative activation functions. Moreover, integrating with the RAD method enhances the model’s flexibility and further improves its performance.

Keywords Physics-informed neural network · Kolmogorov–Arnold network · Residual-based adaptive distribution · Acoustic Scattering

1 Introduction

Deep learning (DL), as a subfield of machine learning, endeavors to design and train multi-layer neural networks that can automatically learn hierarchical feature representations from complex datasets while performing discriminative and predictive tasks through end-to-end learning paradigms [1]. The algorithms utilize real-time monitoring data to capture intricate system dynamics, enhancing operational stability and process efficiency.

Within the scattering domain, DL has witnessed increasing adoption in industrial non-destructive evaluation,

advanced medical imaging, material characterization, and inverse design of functional materials, which underscores its expanding influence and high-impact potential for future advancements in scattering modeling [2, 3]. Chen et al. [4] employed deep neural networks (DNNs) to solve Maxwell’s equations and predict nanoparticle-mediated optical scattering phenomena. Inverse scattering problems [5] for designing acoustic and optical metamaterials have recently been addressed through semi-supervised and unsupervised DL models. However, conventional DL approaches often lack integration with fundamental physical laws, and the trained DNNs cannot intrinsically ensure physically consistent predictions. As a result, these approaches’ performance depended heavily on the quality and quantity of training data [6].

Physics-informed neural networks (PINNs) are considered an efficient scattering simulation technique. Nair et al. [7] developed a PINNs method for exact scattering trajectories. The PINN-based architecture directly enforces the

✉ Ligang Wang
wangligang@hrbeu.edu.cn

¹ School of Mathematical Sciences, Harbin Engineering University, Nantong Street, Harbin 150001, Heilongjiang, China

governing equations describing the underlying physics into the training process. By introducing strong prior knowledge regarding wave propagation and scattering characteristics, this architecture can remarkably reduce computational time and the need for a labeled training dataset. Qu et al. [8] developed a PINNs framework for solving inverse forms of PDEs to analyze acoustic scattering problems, where the inverse forms of PDEs are represented by boundary integral equations (BIEs). Owing to the semianalytic characteristics of BIEs, the proposed method achieves higher precision. Additionally, Zhang et al. [9] proposed coupled PINNs to simulate wave scattering phenomena within unbounded domains and conducted a convergence analysis based on the universal approximation and Sobolev space theories. Beyond PINNs, reinforcement learning (RL) also provides a robust framework for addressing large-scale multi-agent scattering and control challenges [10]. For such complex systems, fault-tolerant control with event-triggered mechanisms further ensures reliability under state constraints and fault risks [11].

Despite the encouraging progress made in PINN-based scattering simulations, existing methods often neglect the inherent spatial continuity in frequency-domain acoustic scattering. However, traditional architectures inherently fail to capture spatial neighborhood dependencies [12], as they lack a "spatial correlation mechanism" that connects information from adjacent spatial points – a function similar to the local receptive field of CNNs in the field of image processing, which enables the integration of neighborhood information [13]. As a result, the impact of boundary conditions or scattering characteristics does not achieve continuous propagation in space, leading the models to maintain satisfactory accuracy only in the near-field close to obstacles, while the prediction error in the far-field increases significantly. To tackle these issues, researchers [14, 15] have proposed some methods utilizing GCN [16] and Sub-sequential [17] architectures, which capture dependencies by explicitly defining relational structures (e.g., the adjacency matrix of GCNs and attention weights of Transformers). These explicit relational structures introduce structural constraints or computational costs, which limit their adaptability, efficiency, and generalization ability in complex scattering scenarios.

To tackle this issue, we introduce PIKAN, an implicitly connected PINNs framework that integrates the global acoustic field modeling capability of the Kolmogorov-Arnold Network (KAN) – a novel neural network architecture based on the Kolmogorov-Arnold representation theorem, which deploys learnable activation functions on edges and uses node-wise summation for mapping [18] – and the local acoustic field feature extraction capability of Sine activation [19]. Unlike explicitly connected

architectures, PIKAN avoids computationally expensive self-attention operations; instead, it leverages KAN-driven plane wave expansion and the efficient plane wave learning capability of Sine activation to ensure consistent and scalable training on different scattering scenarios. By integrating these complementary architectures, PIKAN effectively captures global spatial dependencies and local scattering characteristics in the Helmholtz equation, all while ensuring computational efficiency. The contributions of this study are as follows.

- **New Framework:** For scattering problems, we propose a novel yet intuitive KAN-based framework named PIKAN. This framework equips PINNs with the capability to capture global acoustic field dependencies through plane wave expansion, thereby enhancing the generalization ability and approximation accuracy in effectively solving scattered fields.
- **Novel Activation:** In PIKAN, we introduce a new periodic activation function named Sine. Sine is designed to learn different frequency components of plane waves, enhancing the model's ability to learn local fine-grained features.
- **Extensive Validation:** We assess the effectiveness and generalization capacity of the Sine-based PIKAN model on three types of application scenarios: high-frequency case, irregular scatterer, and multiple scattering. It balances accuracy and computing efficiency better than state-of-the-art baselines and other periodic activations.
- **Combination with RAD method:** The RAD method enhances the Sine-based PIKAN model's ability to capture nonlinear features in key regions, thereby further improving its prediction performance.

The rest of this paper is organized as follows. Section 2 introduces previous related work and establishes the theoretical foundation of the field. Section 3 elaborates on the mathematical formulation of acoustic scattering problems and provides implementation details of fundamental deep learning concepts. Section 4 systematically describes the PIKAN architecture developed for acoustic scattering problems. Section 5 presents comprehensive experimental validation, including performance evaluation on three application scenarios and ablation studies on key hyperparameters. Finally, Section 6 presents conclusions.

2 Related works

PINNs represent a transformative approach at the intersection of machine learning and scientific computing, aimed at solving problems governed by the physical laws described

in PDEs. Raissi et al. [20] pioneered this approach by introducing physical laws into learning, enabling complicated systems to be solved and explored. This work has facilitated its applications in multiple fields, including fluid dynamics, solid mechanics, and geophysics [21]. To enhance capabilities, researchers have designed variants such as Xtended PINN (XPINN), Self-adaptive PINN (SA-PINN), and Bayesian PINN (B-PINN). In terms of theory, Sun et al. [22] combined the Laplace transformation with the final value theorem to prove the asymptotic convergence of errors of the proposed model under different time-varying disturbances. Furthermore, researchers [23–26] conducted stability analysis based on the Lyapunov approach and proved that the key signals within the closed-loop nonlinear system are uniformly ultimately bounded.

In most problems solved by PINNs, the solution is conventionally approximated by the Multi-Layer Perceptron (MLP). The MLP is a fully connected feedforward neural network, and its prevalent use in PINNs stems from the Universal Approximation Theorem, which endows it with universal function approximation capability. Although MLPs have strong approximation capability, they often neglect the physical properties of solutions, leading to failure to converge [27, 28]. Krishnapriyan et al. [29] examines this constraint and determines its underlying cause: the scatter-point loss function of PINNs obviously mismatches the PDE-solving objective, which aims to approximate the solution over a continuous domain, leading to severe pathological curvature of the loss landscape. To tackle the issue, researchers increase the physical constraints by adjusting the distribution of training samples. Sun et al. [30] employed wavelet packet decomposition to extract high and low frequency information from original signal, and subsequently reconstructed the generated high and low frequency information to enhance the diversity of pseudo samples. In this study, we generate the training data distribution based on PDE residuals to address similar challenges of limited distribution variation in complex scattering scenarios.

Recently, KAN has been proposed as an alternative to MLP [31]. Compared with MLP, KAN provides for clearer transparency of their underlying function composition, which makes it easier to customize model components to enhance PDE solutions. Currently, it has been extended to various fields, including PDE solving, image processing, and audio signal classification, among other applications. [18, 32, 33]. Researchers have also developed techniques aimed at enhancing the efficiency of KAN, such as grid sparsification and adaptive state transitions [34]. However, the full potential of KAN-based models in acoustic problems remains largely untapped. Thus, exploring the effective integration of KAN with existing models in acoustic problems is an important and promising research direction.

3 Preliminary

3.1 Problem description of acoustic scattering

The scattering problem caused by plane waves incident on a rigid scatterer is considered [35]. The incoming wave is expressed as $p_i = p_0 e^{-ik \cdot x}$ in which the wave vector $k = 2\pi f \cdot \hat{e}_k / c_s$ is the function of the frequency f and the sound speed c_s , with the vector $\hat{e}_k = [1, 0]$. Figure 1 shows the incoming wave ($p_0 = 1Pa$) specified in a 2D domain $\Omega \subset \mathbb{R}^2$ using the spatial coordinate $x = (x, y)$. The Helmholtz equation governs the problem

$$F(p_s, x) := \nabla^2 p_s(x) + k^2 p_s(x) = 0, x \in \Omega, \quad (1)$$

where F represents the differential operator of the Helmholtz equation, the wavenumber $k = |k|$, and the scattered pressure is denoted by p_s . The Neumann boundary condition on Γ_i is used to describe the rigid boundary of the scatterer S , which can be expressed as

$$B_i(p_s, x) := \frac{\partial p_s(x)}{\partial n} - ike^{-ik \cdot x} = 0, x \in \Gamma_i, \quad (2)$$

where B_i represents the boundary operator of the internal boundary condition and n is the surface normal. To approximate an infinite domain, the Sommerfeld radiation condition is imposed via the first-order DtN boundary condition (termed the absorbing boundary) on the external boundary Γ_e , which can be expressed as

$$B_e(p_s, x) := \frac{\partial p_s(x)}{\partial n} + ikp_s(x) = 0, x \in \Gamma_e, \quad (3)$$

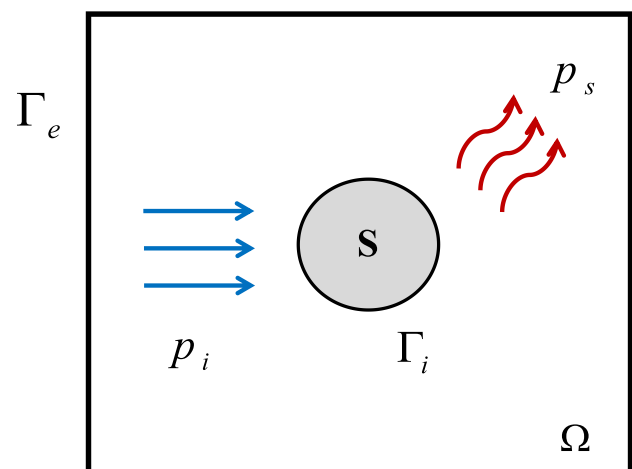


Fig. 1 Schematic diagram of the scattering problem

where B_e represents the boundary operator of the external boundary condition. Equations (1)-(3) explain the scattering problem induced by a rigid scatterer S . The solution to Equation (1) is indicated by complex-valued $p_s = \Re(p_s) + i\Im(p_s)$.

3.2 Physics-Informed Neural Networks (PINNs)

To introduce the PINNs method, we take the following nonlinear differential equation as an example.

$$\begin{aligned}\mathcal{F}_\eta(u(x)) &= \varphi(x), \quad x \in \Omega, \\ \mathcal{B}_\eta^k(u(x)) &= b^k(x), \quad x \in \Gamma^k \subseteq \partial\Omega,\end{aligned}\quad (4)$$

where $\mathcal{F}_\eta(\cdot)$ is a nonlinear differential operator, and the subscript η is related parameters, u is the solution of the differential equation, φ is the source term, $\{\mathcal{B}_\eta^k\}_{k=1}^{n_\Gamma}$ is a set of n_Γ boundary operators and b is the boundary term.

When employing PINNs to solve the differential equation, the exact solution $u(x)$ is approximated by a neural network $\hat{u}(x; \theta)$, parameterized by the set θ . By leveraging the automatic differentiation technique, we can efficiently calculate arbitrary-order derivatives of $\hat{u}(x; \theta)$. The technique is essential for calculating the loss function $\mathcal{L}(\theta)$ in PINNs. The total loss $\mathcal{L}_\theta$ is defined as the weighted sum of residual loss $\mathcal{L}_F$ and boundary loss $\mathcal{L}_B^k$.

$$\mathcal{L}_F(\theta) = \frac{1}{N_r} \sum_{j=1}^{N_r} \|\mathcal{F}_\eta(\hat{u}(x_j^r; \theta)) - \varphi_j\|^2, \quad (5)$$

$$\mathcal{L}_B^k(\theta) = \frac{1}{N_b^k} \sum_{l=1}^{N_b^k} \|\mathcal{B}_\eta^k(\hat{u}(x_l^b; \theta)) - b_l^k\|^2, \quad (6)$$

$$\mathcal{L}(\theta) = \omega_F \mathcal{L}_F(\theta) + \sum_{k=1}^{n_\Gamma} \omega_B^k \mathcal{L}_B^k(\theta), \quad (7)$$

where N_r and N_b^k denote the numbers of residual points and boundary points, respectively, and ω_F, ω_B^k are weighting coefficients of the residual loss $\mathcal{L}_F$ and boundary loss $\mathcal{L}_B^k$ in the total loss $\mathcal{L}_\theta$.

3.3 Residual-Based Adaptive Distribution (RAD)

In the construction of discrete loss functions for PINNs, the selection of sampling points is of great significance to the training of PINNs. However, because it is challenging to obtain a specific distribution related to solution characteristics using a traditional uniform sampling method in complex scenarios,

the collected datasets will be unbalanced, which will greatly affect the accuracy of the approximate solution [6].

To tackle the above mentioned challenges, researchers have focused on developing adaptive sampling strategies [36]. Motivated by the adaptive mesh refinement in the Finite Element Method (FEM), Wu et al. [37] proposed the residual-based adaptive distribution (RAD) method for PINNs, which is an adaptive non-uniform sampling method. All residual points are resampled using a probability density function $\rho(x)$ proportionate to the PDE residual (Algorithm 1). Specifically, for any point x , we first compute the PDE residual $\varepsilon(x) = \|\mathcal{F}_\eta(\hat{u}(x; \theta)) - \varphi_j\|$, and then compute a probability as

$$\rho(x) \propto \frac{\varepsilon^k(x)}{\mathbb{E}[\varepsilon^k(x)]} + c, \quad (8)$$

where $k \geq 0$ and $c \geq 0$ serve as hyperparameters to regulate the sampling density. Numerical integration like Monte Carlo can approximate $\mathbb{E}[\varepsilon^k(x)]$. The number of rounds R used by RAD is a hyperparameter. RAD sampling ensures sample point randomization while balancing model training in problematic and other domains.

```

1: Sample the initial residual points  $\mathcal{T}$  via uniform sampling,  $|\mathcal{T}| = N_r$ ;
2: for round = 0, 1, 2, ...,  $R$  do
3:   Train the PINNs for a certain number of iterations on the residual set  $\mathcal{T}$ ;
4:   Sample a set of dense points  $\mathcal{S}$  via uniform sampling,  $(|\mathcal{S}| \gg |\mathcal{T}|)$ ;
5:   Compute  $\rho(x)$  for the points in  $\mathcal{S}$ ;
6:   Define a probability mass function  $\tilde{\rho}(x) = \frac{\rho(x)}{A}$ ,  $A = \sum_{x \in \mathcal{S}} \rho(x)$ ;
7:   Sample a subset  $\mathcal{T}$  of points from  $\mathcal{S}$  according to  $\tilde{\rho}(x)$ .
8: end for

```

Algorithm 1 Residual-based adaptive distribution (RAD)

4 Methodology

4.1 Kolmogorov-Arnold networks

The Kolmogorov-Arnold Representation Theorem (KART), proposed by Kolmogorov and Arnold, states that any multivariate continuous function on a bounded domain can be expressed as a finite composition of univariate continuous functions and addition [31]. Specifically, for a continuous function $f: [0, 1]^n \rightarrow \mathbb{R}$,

$$f(x) = f(x_1, \dots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right), \quad (9)$$

where $\phi_{q,p}$ and Φ_q denote inner and outer functions, respectively. According to Equation (9), theoretically, any high-dimensional complex function can be represented by two layers of univariate functions. However, these functions can lack smoothness and even exhibit fractal characteristics, and for neural networks, such complex functions cannot be learned. The innovation of KAN lies in breaking through the limitation of the two-layer functions in the KART and designing the network structure as a multi-layer [18]. KAN uses learnable activation functions on edges, unlike standard MLP which employs fixed functions on neuron (see Fig. 2). Next, we introduce the mathematical details of KAN. An L -layers KAN can be expressed as

$$[n_0, n_1, \dots, n_L], \quad (10)$$

where n_i represents the neuron number in the i^{th} layer. The i th neuron in the l^{th} layer is defined as (l, i) , and its activation value as $x_{l,i}$. The activation function that links (l, i) and $(l+1, j)$ is denoted as

$$\phi_{l,j,i}, \quad l = 0, \dots, L-1, \quad i = 1, \dots, n_l, \quad j = 1, \dots, n_{l+1}. \quad (11)$$

The pre-activation of $\phi_{l,j,i}$ is $x_{l,i}$, and the post-activation is $\tilde{x}_{l,j,i} \equiv \phi_{l,j,i}(x_{l,i})$. All incoming post-activations make up the neuron's $(l+1, j)$ activation value:

$$x_{l+1,j} = \sum_{i=1}^{n_l} \tilde{x}_{l,j,i} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_{l,i}), \quad j = 1, \dots, n_{l+1}. \quad (12)$$

This is expressed in matrix form as

$$x_{l+1} = \underbrace{\begin{pmatrix} \phi_{l,1,1}(\cdot) & \phi_{l,1,2}(\cdot) & \dots & \phi_{l,1,n_l}(\cdot) \\ \phi_{l,2,1}(\cdot) & \phi_{l,2,2}(\cdot) & \dots & \phi_{l,2,n_l}(\cdot) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{l,n_{l+1},1}(\cdot) & \phi_{l,n_{l+1},2}(\cdot) & \dots & \phi_{l,n_{l+1},n_l}(\cdot) \end{pmatrix}}_{\Phi_l} x_l, \quad (13)$$

where Φ_l is the function matrix of the l^{th} layer. For a L -layers KAN, given an input vector $\mathbf{x} \in \mathbb{R}^n$, the output of KAN is

$$\text{KAN}(\mathbf{x}) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \dots \circ \Phi_1 \circ \Phi_0)\mathbf{x}. \quad (14)$$

Meanwhile, the original KART (Equation 9) corresponds to a 2-layer KAN with shape $[n, 2n+1, 1]$.

4.2 Plane wave expansion of scattered acoustic field

Before developing the PIKAN model for acoustic scattering problems, it is necessary to clarify the inherent advantages of using KAN as its underlying architecture. According to the Plane Wave Expansion method (PWE), we can regard

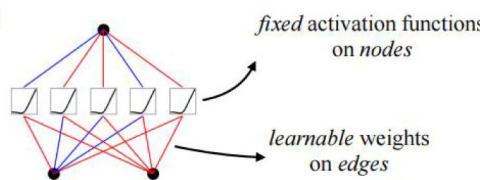
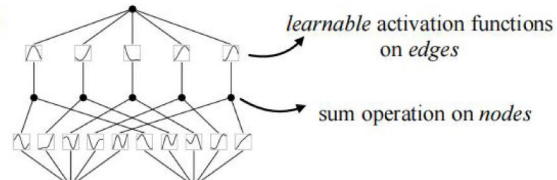
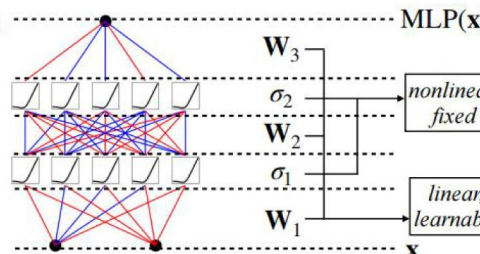
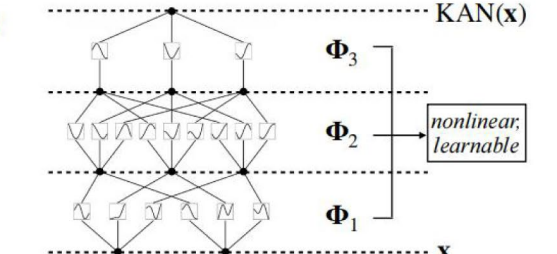
Model	MLP	KAN
Theorem	Universal Approximation Theorem	Kolmogorov-Arnold Representation Theorem
Formula (Shallow)	$f(\mathbf{x}) \approx \sum_{i=1}^{N(\epsilon)} a_i \sigma(\mathbf{w}_i \cdot \mathbf{x} + b_i)$	$f(\mathbf{x}) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$
Model (Shallow)	(a) 	(b) 
Formula (Deep)	$\text{MLP}(\mathbf{x}) = (\mathbf{W}_3 \circ \sigma_2 \circ \mathbf{W}_2 \circ \sigma_1 \circ \mathbf{W}_1)(\mathbf{x})$	$\text{KAN}(\mathbf{x}) = (\Phi_3 \circ \Phi_2 \circ \Phi_1)(\mathbf{x})$
Model (Deep)	(c) 	(d) 

Fig. 2 Comparison between MLP and KAN architectures [18]

the frequency-domain scattered acoustic field that satisfies the Helmholtz equation as the superposition of plane waves in different directions [3, 38]. Specifically, the 2D scattered acoustic field p_s can be expressed

$$\begin{aligned} p_s(x) &= \int_{\Omega} p_{s,k}(x) d\Omega(k) \\ &= \int_{\Omega} A(k) \cdot e^{ik \cdot x} d\Omega(k), \end{aligned} \quad (15)$$

where $A(k)$ and $e^{ik \cdot x}$ serve as the amplitude coefficient and phase factor of the plane wave $p_{s,k}(x)$, respectively. $k = (k_x, k_y) = (k \cos \theta, k \sin \theta)$ is its wave vector. The polar coordinate representation of Equation (15) is as follows

$$p_s(x) = \int_0^{2\pi} A(\theta) \cdot e^{i[k(x \cos \theta + y \sin \theta) + \varphi(\theta)]} d\theta. \quad (16)$$

This integral can be discretized using Riemannian sums in a finite set of directions $\theta = \{\theta_1, \theta_2, \dots, \theta_N\}$ where $\theta_n = 2\pi(n-1)/N$ (i.e., uniformly discretized $0 \sim 2\pi$ into N directions).

$$p_s(x) \approx \sum_{n=1}^N A_n \cdot e^{i[k(x \cos \theta_n + y \sin \theta_n) + \varphi_n]}, \quad (17)$$

where we make the replacements $\int_0^{2\pi} \rightarrow \sum_n$, $A(\theta)d\theta \rightarrow A_n$ and $\theta, \varphi(\theta) \rightarrow \theta_n, \varphi_n$. Theoretically, when N approaches infinity, the series can achieve an arbitrarily accurate approximation of $p_s(x)$. In practice, most solutions of scattered acoustic fields contain only a finite number of plane waves, then the approximation of the true solution can also be achieved using finite parameters N .

Equation (17) can be further expanded, and we take the real part as an example:

$$\begin{aligned} Re(p_s(x)) &\approx \sum_{n=1}^N A_n \cdot \cos((k \cos \theta_n x + \varphi_{n,1}) + (k \sin \theta_n y + \varphi_{n,2})) \\ &= \sum_{n=1}^N A_n \cdot [\cos(X) \cos(Y) - \sin(X) \sin(Y)], \end{aligned} \quad (18)$$

where $X = k \cos \theta_n x + \varphi_{n,1}$, $Y = k \sin \theta_n y + \varphi_{n,2}$. The product in Equation (18) can be obtained by addition.

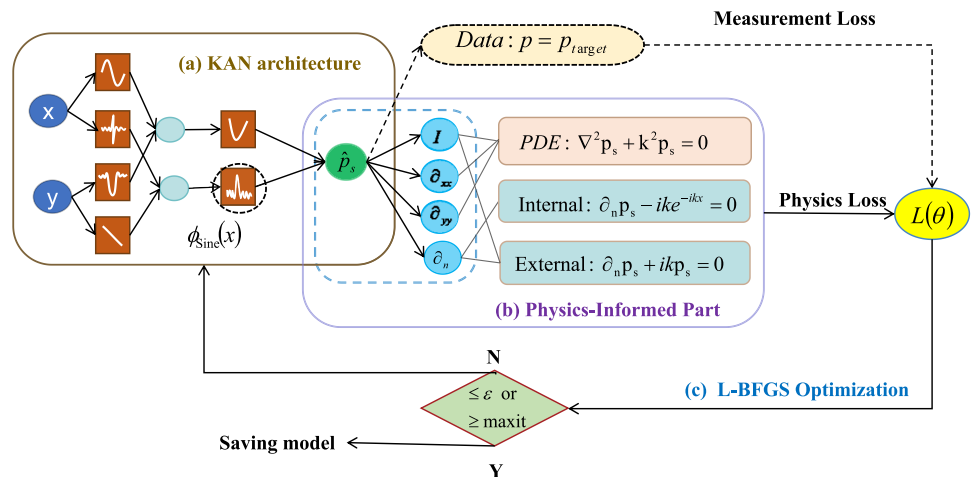
To sum up, we find that the scattered acoustic field p_s can finally be decomposed into a superposition of one-dimensional functions, which is naturally compatible with the one-dimensional function decomposition property of KAN. From a physical perspective, the learning of the scattered acoustic field p_s boils down to the learning of one-dimensional functions at different frequencies. From a model perspective, the width and depth of KAN correspond respectively to the number of plane waves needed to expand the scattered acoustic field, and the computational complexity to calculate plane waves in a single direction.

4.3 Development of PIKAN for acoustic scattering

Based on the findings from the Section 4.2, we adopt KAN as the underlying architecture to develop the PIKAN model for acoustic scattering problems.

The architecture of the PIKAN is illustrated in Fig. 3. First, the architecture takes an input coordinate (x, y) and passes it through a series of KAN layers to obtain an output $\hat{p}_s$. Next, the gradients of $\hat{p}_s$ are calculated using automated differentiation with regards to (x, y) . This embeds the physics-informed part into the network, resulting in a physically consistent loss function. Moreover, the proposed model is exclusively governed by physical principles, eliminating the requirement for labeled data during its training phase. Finally, the L-BFGS optimizer is employed to perform backpropagation to optimize the model until the (loss/gradient) falls below the threshold or the maximum number of iterations is reached.

Fig. 3 The model architecture of PIKAN



To enforce Equations (1)-(3), the physics-driven loss function is defined as follows

$$\begin{aligned}\mathcal{L}_F(\theta) &= \frac{1}{N_r} \sum_{j=1}^{N_r} \|F(\hat{p}_s, \mathbf{x}_r^j; \theta)\|^2 \\ &= \frac{1}{N_r} \sum_{j=1}^{N_r} \|\nabla^2 \hat{p}_s(\mathbf{x}_r^j) + k^2 \hat{p}_s(\mathbf{x}_r^j)\|^2,\end{aligned}\quad (19)$$

$$\begin{aligned}\mathcal{L}_{B_i}(\theta) &= \frac{1}{N_{b_i}} \sum_{j=1}^{N_{b_i}} \|B_i(\hat{p}_s, \mathbf{x}_{b_i}^j; \theta)\|^2 \\ &= \frac{1}{N_{b_i}} \sum_{j=1}^{N_{b_i}} \left\| \frac{\partial \hat{p}_s(\mathbf{x}_{b_i}^j)}{\partial \mathbf{n}} - i k e^{-i \mathbf{x}_{b_i}^j} \right\|^2,\end{aligned}\quad (20)$$

$$\begin{aligned}\mathcal{L}_{B_e}(\theta) &= \frac{1}{N_{b_e}} \sum_{j=1}^{N_{b_e}} \|B_e(\hat{p}_s, \mathbf{x}_{b_e}^j; \theta)\|^2 \\ &= \frac{1}{N_{b_e}} \sum_{j=1}^{N_{b_e}} \left\| \frac{\partial \hat{p}_s(\mathbf{x}_{b_e}^j)}{\partial \mathbf{n}} + i k \hat{p}_s(\mathbf{x}_{b_e}^j) \right\|^2,\end{aligned}\quad (21)$$

where N_r , N_{B_i} and N_{B_e} denote the numbers of residual points, internal boundary points and external boundary points, respectively. Next, we define the total loss $\mathcal{L}_{total}(\theta)$ as:

$$\mathcal{L}(\theta) = \omega_F \mathcal{L}_F(\theta) + \omega_{B_i} \mathcal{L}_{B_i}(\theta) + \omega_{B_e} \mathcal{L}_{B_e}(\theta), \quad (22)$$

where ω_N , ω_{B_i} , and ω_{B_e} , denote the weighting coefficients of the residual loss, internal boundary loss, and external boundary loss, respectively. Additionally, for multiple scattering, the internal boundary loss is the sum of the losses on the boundary of all scatterers.

4.4 Activation

In KAN, the activation function $\phi(x)$ is generated by B-spline basis functions. However, their performance in PIKAN may not always be effective. Research by Shukla et al. [32] and Zavareh et al. [39] suggests that the application of B-spline activation within PIKAN results in lower efficiency. This is since B-splines require recursive computation, making them poorly suited for the parallel computing architecture of modern GPUs. And the piecewise nature of B-splines hinders derivative calculation, which is important for PIKAN's performance. Recent research has explored various KAN variants with alternative activation functions, such as rKAN [40], Sinc-KAN [41], and Fourier-KAN [19], which will be discussed in Section 5.3

Any sufficiently well-behaved/smooth one dimensional function $f: \mathbb{R} \rightarrow \mathbb{R}$ can be expressed in terms of a *Fourier transform* $\tilde{f}: \mathbb{R} \rightarrow \mathbb{C}$ w.r.t. a continuous phase space of frequencies ω :

$$\begin{aligned}f(x) &= \int_{\mathbb{R}} d\omega \tilde{f}(\omega) e^{i\omega x} \\ &= \text{Re} \left(\int_{\mathbb{R}} d\omega \tilde{f}(\omega) e^{i\omega x} \right) \\ &= \int_{\mathbb{R}} d\omega A(\omega) \cos(\omega x + \psi'(\omega)) \\ &= \int_{\mathbb{R}} d\omega A(\omega) \sin(\omega x + \psi(\omega)),\end{aligned}\quad (23)$$

where $A(\omega)$ and $\psi(\omega) = \psi'(\omega) - \frac{\pi}{2}$ are real functions. The above integral can be discretized over a finite set of frequencies $\Omega = \{\omega_1, \omega_2, \dots, \omega_G\}$ in which cardinality G is the grid size. The function ϕ_{Sine} is proposed as an ansatz for $f(x)$.

$$\phi_{Sine}(x) = \sum_{i=1}^G A_i \cdot \sin(\omega_i x + \psi_i). \quad (24)$$

In the PIKAN model we proposed, the Sine function (Equation (24)) [19] is employed as the activation function, where $\theta = \{\omega_i, \psi_i, A_i\}_{i=1}^G$ are learnable parameters. The Sine activation function of the PIKAN model naturally aligns with the inherent mathematical formulation of the scattered field, specifically the trigonometric superposition resulting from plane wave expansion, enhancing the model's ability to learn local features. To a certain extent, G can be understood as the number of frequency components covered by a single plane wave. Additionally, the Sine function features a simple computation process, making it highly suitable for parallel computing and thereby enhancing the computational efficiency.

5 Experiments

5.1 Setup

Our experiments cover three typical scattering scenarios: high-frequency cases, irregular scatterer, and multiple scattering. The three challenging scenarios are often used as benchmarks for scattering simulations. Meanwhile, we choose vanilla PINN [20], PINN-GCN [16], PINNsFormer [17], and PINNMamba [14] as baseline models.

We uniformly sampled $N_{b_i} = N_{b_e} = 1000$ internal and external boundary points and $N_r = 10000$ residual points

Table 1 Model-specific hyperparameter settings used in experiments

Model	Hyperparameter Type	Value
PINN	Network depth	4
	Network width	512
PINN-GCN	Network depth	4
	Network width	256
PINNsFormer	Encoder / Decoder	1 / 1
	Embedding size	32
	Attention heads	2
	MLP hidden width	512
	Sequence length k	5
	Sequence interval Δt	1×10^{-4}
	Sequence interval Δt	1×10^{-2}
PINNMamba	Encoder layers	1
	Embedding size	32
	Δ , B, C width	8
	MLP hidden width	512
	Sequence length k	7
	Sequence interval Δt	1×10^{-2}
	Sequence interval Δt	1×10^{-2}
PIKAN	Network architecture	[2,5*[10],2]
	Grid size	10

to build the training dataset. For testing, we employed a 101×101 mesh within the residual domain. The FEM method is employed as a ground truth solution to simulate the scattering problem. To ensure fairness in comparison, we train PIKAN and all the baseline models for 1000 epochs with the L-BFGS optimizer. To maximize baseline model performance, we use their authors' suggested hyperparameter settings, which are detailed in Table 1. The weights of loss terms $[\omega_F, \omega_{B_i}, \omega_{B_e}]$ are set to [1, 10, 1] for all three scenarios, as we find that strengthening the internal boundary conditions can lead to better convergence. All models are implemented in PyTorch 2.4.0 and trained on a single NVIDIA GeForce RTX 4070 Ti SUPER.

To evaluate the model performance, we adopt relative Mean Absolute Error (rMAE, or relative ℓ_1 error) and relative Root Mean Square Error (rRMSE, or relative ℓ_2 error) as evaluation metrics:

$$\text{rMAE}(\hat{p}) = \frac{\sum_{n=1}^N |\hat{p}_s(x_n) - p_s(x_n)|}{\sum_{n=1}^N |p_s(x_n)|}, \quad (25)$$

$$\text{rRMSE}(\hat{p}) = \sqrt{\frac{\sum_{n=1}^N (\hat{p}_s(x_n) - p_s(x_n))^2}{\sum_{n=1}^N (p_s(x_n))^2}}, \quad (26)$$

where N is the number of testing points, $p_s(x)$ is the ground truth, and $\hat{p}_s(x)$ is the model's approximation.

5.2 Performance comparison

Table 2 presents the quantitative performance comparison of PIKAN and other baseline models on three scattering scenarios, including the high-frequency case, irregular scatterer, and multiple scattering. The results showed that the PIKAN model consistently performs better than other baselines, achieving the lowest training loss and test errors.

Specifically, as shown in Fig. 4, for high-frequency scattering, PIKAN effectively performs global acoustic field modeling by learning plane waves in different directions, while other models ignore, to varying degrees, the inherent essential physical features in acoustic scattering. For problems as intrinsically difficult to optimize, as shown in Fig. 5, PIKAN can further optimize at the boundary, resulting in a lower error than PINN-GCN and PINNMamba for irregular scattering. The superposition principle dictates that the scattering field at any point equals the sum of the fields generated by each scatterer. As shown in Fig. 6, PIKAN learns the scattered field of each scatterer individually, thus achieving significantly higher accuracy. These results verify the adaptability and innovation of the PIKAN model in extensive acoustic scattering problems.

Table 3 presents the parameters and training time of all models on three application scenarios. The results showed that PIKAN balances accuracy and computational efficiency. Specifically, compared with PINNMamba, with only 4.6% of the parameter count, PIKAN improves accuracy by 61.4% and accelerates training by over 79.1%. Compared with PINN, which is known for its efficiency, on the multiple scattering, PIKAN only increases training time by 26.3%, while improving accuracy by 83.8% and reducing the parameter count by 97.5%.

Table 2 Prediction errors for different baseline models on high-frequency, irregular scatterer, and multiple scattering problems

Model	High-frequency			Irregular scatterer			Multiple scattering		
	Loss	rMAE	rMSE	Loss	rMAE	rMSE	Loss	rMAE	rMSE
PINN	0.0237	0.7628	0.7632	0.1191	0.9935	0.9859	0.0316	0.4339	0.4343
PINN-GCN	0.0758	0.8159	0.8141	0.0106	0.1641	0.1683	0.0230	0.1143	0.1166
PINNsFormer	0.0067	0.3439	0.3458	0.0298	0.5987	0.5949	0.0230	0.3346	0.3349
PINNMamba	0.0006	0.0469	0.0629	0.0250	0.2409	0.2426	3e-5	0.1265	0.1458
PIKAN	0.0003	0.0395	0.0563	0.0015	0.0925	0.0935	2e-5	0.0616	0.0704

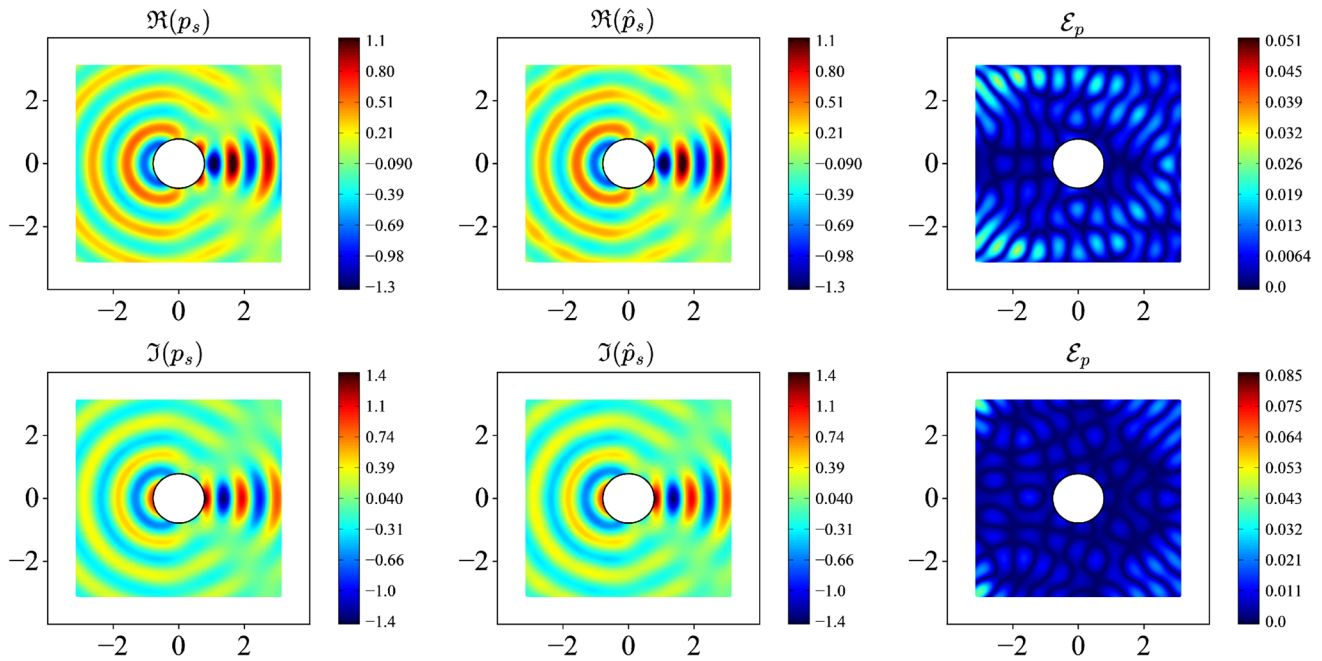


Fig. 4 The ground truth solution (left), prediction (middle), and absolute error (right) of PIKAN on high-frequency scattering scenarios

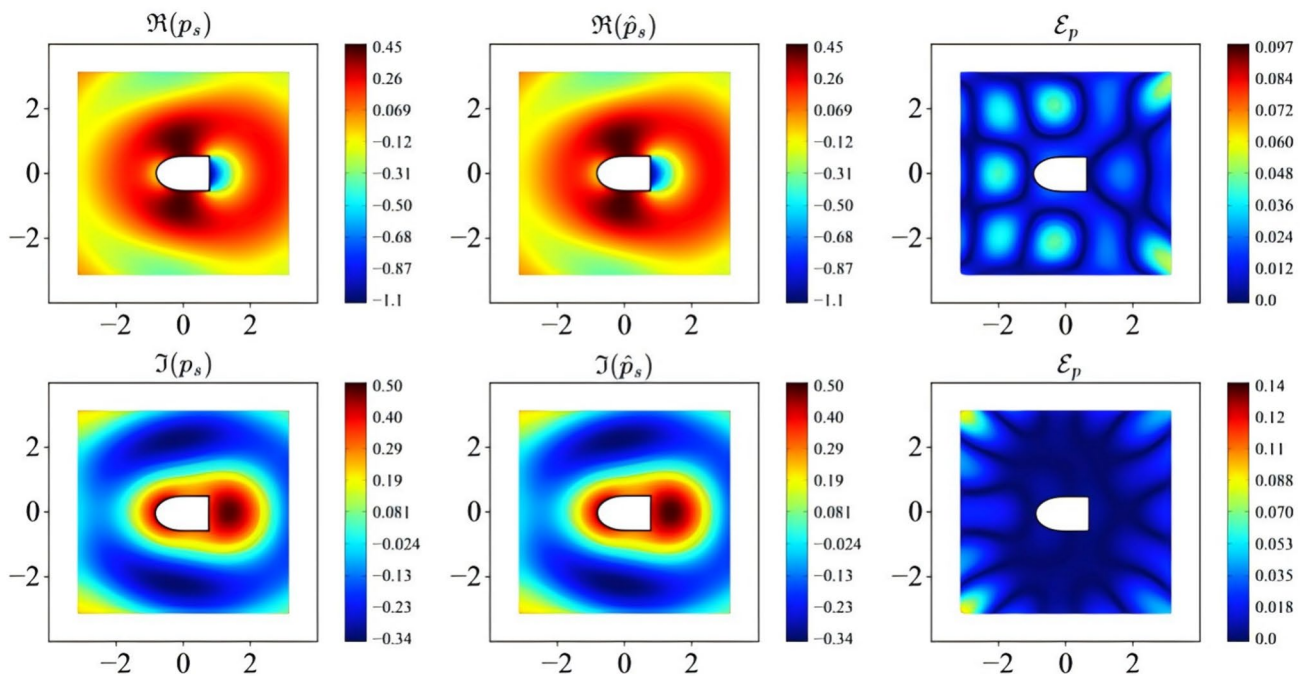


Fig. 5 The ground truth solution (left), prediction (middle), and absolute error (right) of PIKAN on irregular scatterer scenarios

5.3 Activation comparison

To verify the advantages of Sine activation in acoustic scattering problems, we compared the performance differences of PIKAN variants constructed with different activation functions, including B-Spline [18], Rational [40], Sinc [41], Wavelet [39], and Fourier activation functions [19]. For a

fair comparison, we employed the same network architecture and training scheme as in the previous section. For the Sinc activation, the single-layer architecture recommended by the original author was adopted. For different PIKAN variants, we maintain approximately close numbers of parameters to highlight the advantages of Sine activation from its ability to capture the trigonometric superposition

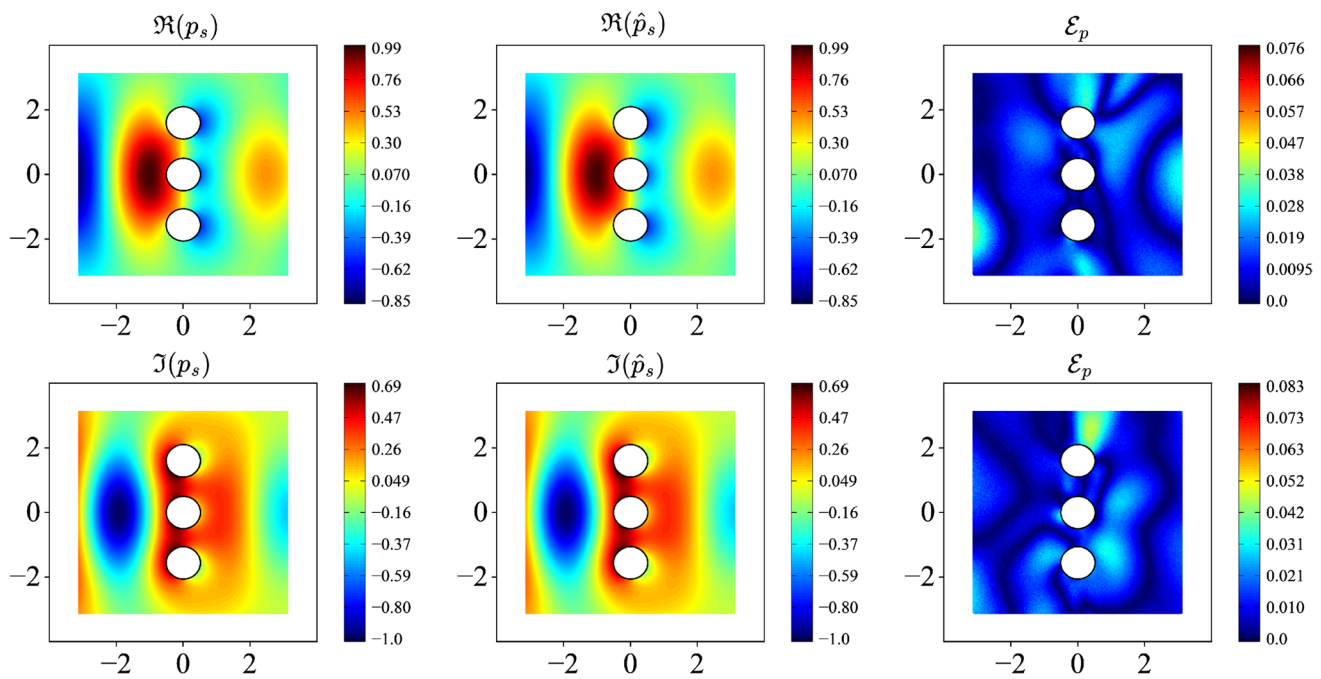


Fig. 6 The ground truth solution (left), prediction (middle), and absolute error (right) of PIKAN on multiple scattering scenarios

Table 3 The model parameters and training time in Performance Comparison

Model	Params	High-frequency		Irregular scatterer		Multiple scattering	
		Time ¹	Time ²	Time ¹	Time ²	Time ¹	Time ²
PINN	527,873	0.29 s/it	15 min	0.42 s/it	22 min	0.37 s/it	19 min
PINN-GCN	132,864	0.58 s/it	26 min	0.48 s/it	25 min	0.45 s/it	25 min
PINNsFormer	454,081	1.82 s/it	73 min	2.36 s/it	91 min	1.87 s/it	110 min
PINNMamba	286,277	1.99 s/it	86 min	2.97 s/it	124 min	2.24 s/it	116 min
PIKAN	13,200	0.67 s/it	28 min	0.89 s/it	26 min	0.69 s/it	24 min

¹ The time per iteration for the model (s/it)

² The total wall-clock time to convergence for the model (Minutes)

Table 4 Activation-specific hyperparameter settings used in experiments

Activation	Hyperparameter Type	Value
B-Spline	Spline order	3
	Grid intervals	25
Rational	Degree of Polynomial (m, n)	(15,15)
	Base step size h_0	10
Sinc	Number of step sizes M	6
	Sinc degree N_{degree}	100
Fourier	Grid size g	15
Wavelet	Wavelet Type	Morlet
	Center frequency ω_0	5
Sine	Grid size G	10

of acoustic fields rather than relying solely on model over-parameterization. Detailed hyperparameters are provided in Table 4.

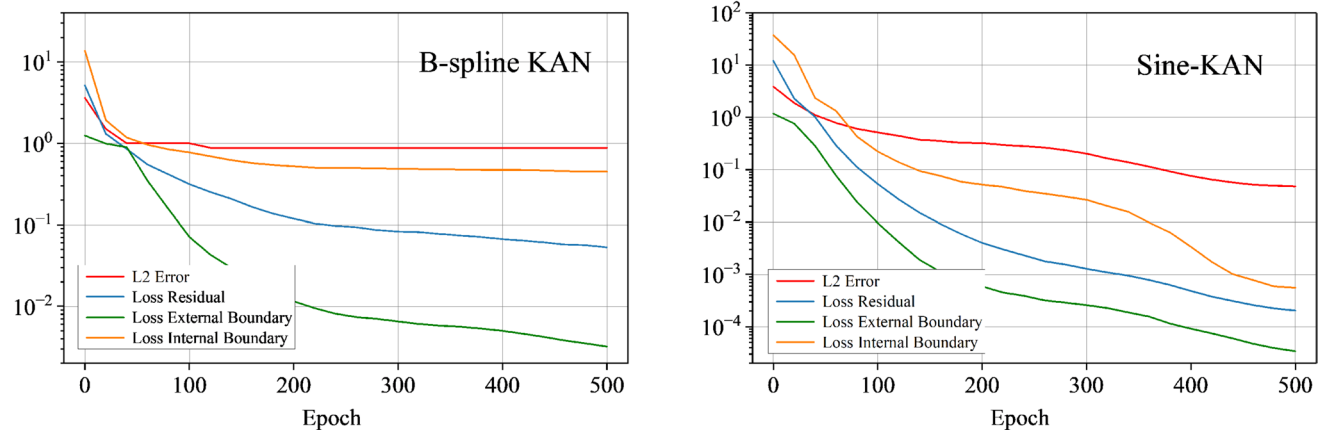
Table 5 presents the quantitative performance comparison of Sine activation and other activations on three scattering scenarios, including the high-frequency case, irregular

scatterer, and multiple scattering. The results demonstrate that periodic activation functions significantly outperform other activation functions concerning solution accuracy. For example, Sine activation allows the model converge to optimal outcomes, while the B-Spline activation severely underperforms in accuracy.

As illustrated in Fig. 7, we analyze the loss curves of B-Spline and Sine activation on high-frequency scattering. During the training process, the rMSE of B-spline KAN does not decrease as $\mathcal{L}_F$ and $\mathcal{L}_{B_e}$ do. This indicates that B-spline KAN is trapped in an over-smoothing solution. Compared with B-spline KAN, the loss reduction process of Sine-KAN shows a high degree of consistency with its error reduction process. This suggests that Sine-KAN does not tend to fall into a local optimum of oversimplified patterns. Instead, it tends to exhibit patterns that are consistent with the original PDEs. The consistency can be assigned to its Sine-added local plane wave learning, which allows PIKAN to perform plane wave expansion more effectively.

Table 5 Prediction errors for different activation functions on high-frequency, irregular scatterer, and multiple scattering problems

Activation	High-frequency			Irregular scatterer			Multiple scattering		
	Loss	rMAE	rMSE	Loss	rMAE	rMSE	Loss	rMAE	rMSE
B-Spline	0.5761	0.8805	0.8807	1.1354	1.2974	1.2868	0.9967	0.9999	0.9999
Rational	0.2294	0.7976	0.7957	0.3139	0.9685	0.9744	0.3395	0.9937	0.9978
Sinc	0.0422	0.3771	0.3974	0.0638	0.7572	0.7685	0.0998	0.6249	0.6182
Fourier	0.0004	0.0487	0.0599	0.0078	0.1504	0.1567	3e-4	0.0862	0.0993
Wavelet	0.0006	0.0511	0.0610	0.0069	0.1303	0.1258	3e-4	0.0698	0.0782
Sine	0.0003	0.0395	0.0563	0.0015	0.0925	0.0935	2e-5	0.0616	0.0704

**Fig. 7** Loss and ℓ_2 -error Curve w.r.t Training Iteration**Table 6** The model training time in Activation Comparison

Model	High-frequency		Irregular scatterer		Multiple scattering	
	Time ¹	Time ²	Time ¹	Time ²	Time ¹	Time ²
B-Spline	2.86 s/it	116 min	2.32 s/it	103 min	2.41 s/it	111 min
Rational	0.98 s/it	49 min	1.03 s/it	45 min	1.12 s/it	47 min
Sinc	0.57 s/it	25 min	0.55 s/it	22 min	0.62 s/it	24 min
Fourier	0.61 s/it	26 min	0.88 s/it	33 min	0.73 s/it	36 min
Wavelet	0.56 s/it	27 min	0.59 s/it	28 min	0.66 s/it	33 min
Sine	0.67 s/it	23 min	0.89 s/it	25 min	0.69 s/it	24 min

¹ The time per iteration for the model (s/it)² The total wall-clock time to convergence for the model (Minutes)

Among the periodic activation functions, Sine activation achieves the lowest error on three application scenarios. Wav-KAN leverages wavelet transforms to capture high-frequency and low-frequency components of the data, thus achieving suboptimal error. While the Sinc activation is not effective in acoustic scattering problems, owing to the oscillations caused by the inaccuracy of derivatives. Furthermore, the Sine activation achieves higher accuracy with a shorter time (Table 6). Specifically, in comparison to Wavelet activation, Sine activation improves accuracy by 25.6% and decreases training by more than 11%. For the multiple scattering problems, the training process is accelerated by 27%. These results show that Sine activation balances accuracy with computational cost.

5.4 Combination with RAD method

In acoustic scattering problems caused by scatterers, there are many challenges in capturing the nonlinear characteristics near the scatterer boundary. For example, nonlinear effects generate high-frequency harmonics, whose wavelengths may be as short as the order of the grid size. If the grid cannot resolve these short waves, it will lead to spurious energy feedback (such as aliasing errors), and thus trigger the spatial propagation of errors. The RAD method guides computational resources to concentrate on complex regions requiring accurate approximation by dynamically adjusting the sampling density, thereby achieving targeted optimization.

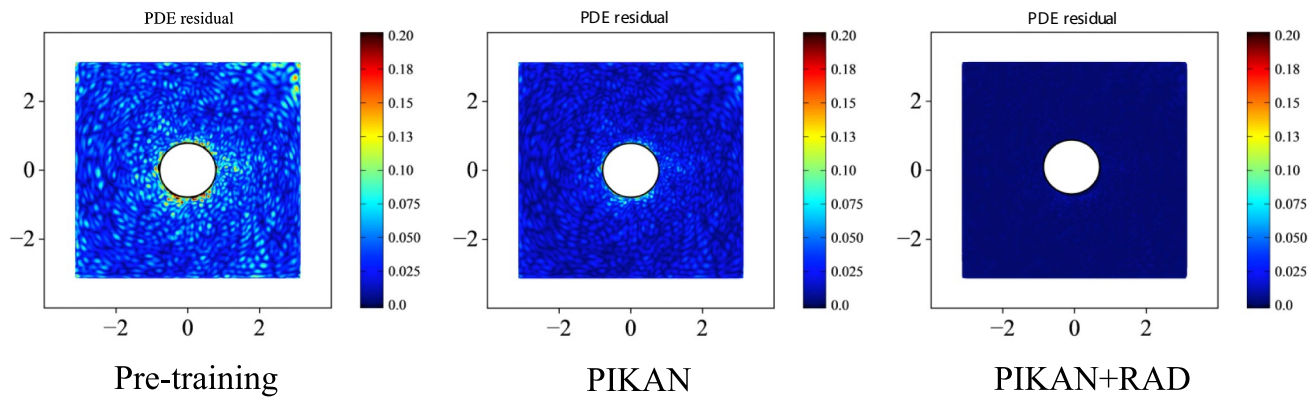


Fig. 8 PDE residual of pressure prediction of high-frequency scattering problem

Table 7 Prediction errors for different network architectures on high-frequency, irregular scatterer, and multiple scattering problems

Architecture	High-frequency			Irregular scatterer			Multiple scattering		
	Loss	rMAE	rMSE	Loss	rMAE	rMSE	Loss	rMAE	rMSE
$[2, 3 * [5], 2]$	0.0038	0.1865	0.1803	0.0344	0.5993	0.5869	$7e-4$	0.2667	0.2721
$[2, 3 * [10], 2]$	0.0041	0.1047	0.1039	0.0095	0.1184	0.1230	$3e-5$	0.1003	0.0981
$[2, 5 * [10], 2]$	0.0003	0.0395	0.0563	0.0015	0.0925	0.0935	$2e-5$	0.0616	0.0704
$[2, 5 * [15], 2]$	0.0006	0.0542	0.0695	0.0011	0.0862	0.0869	$2e-5$	0.0716	0.0789

We first explore the feasibility and performance of its combination with the PIKAN model. The combination of the RAD method in PIKAN follows the description in Section 3.2. We empirically set the RAD parameters $(k, c) = (3, 10)$, sampling rounds $R = 1$, and the number of residual points $|\mathcal{T}| = 10000$, other parameters remain consistent with the previous. The results are illustrated in Fig. 8. The experimental results show that compared with uniform sampling, the combination with the RAD method can more effectively capture the nonlinear characteristics near scatterer boundaries and control the propagation of errors.

To theoretically analyze the error convergence speed of the RAD method in acoustic scattering problems, we initially formulate an error function to quantitatively assess the total error.

$$E_R(t) = \frac{1}{|\mathcal{T}|} \int_{\mathcal{T}} |\hat{p}_{s,R}(x, t) - p_s(x)| dx, \quad (27)$$

where $\hat{p}_{s,R}(x, t)$ is the approximate solution after t iterations of the model in the R -th round. PIKAN with uniform sampling reduce error according to a power law:

$$E_R(t) = \mathcal{O}(t^{-\alpha}). \quad (28)$$

Note that t is the number of training iterations in the R -th round, whereas α is a constant based on the network structure and residual region characteristics. The error decays

uniformly over the domain, while challenging regions are approximated less accurately [36].

Uniform sampling methods may overemphasize the global region while neglecting local details. By adjusting (k, c) , RAD allows more flexibility in refining large-residue locations and covering a broad range. This results in a faster overall convergence speed of PIKAN. The error evolution can be expressed as

$$E_R(t) = \mathcal{O}(t^{-\alpha_R}), \quad (29)$$

where α_R is the decay rate with RAD sampling in the R -th round, which depends on the degree of matching between (k, c) and the residual regions. Typically, $\alpha_R > \alpha$, and we provide a sensitivity analysis of these parameters as well as guidelines for their selection in Section 5.5.

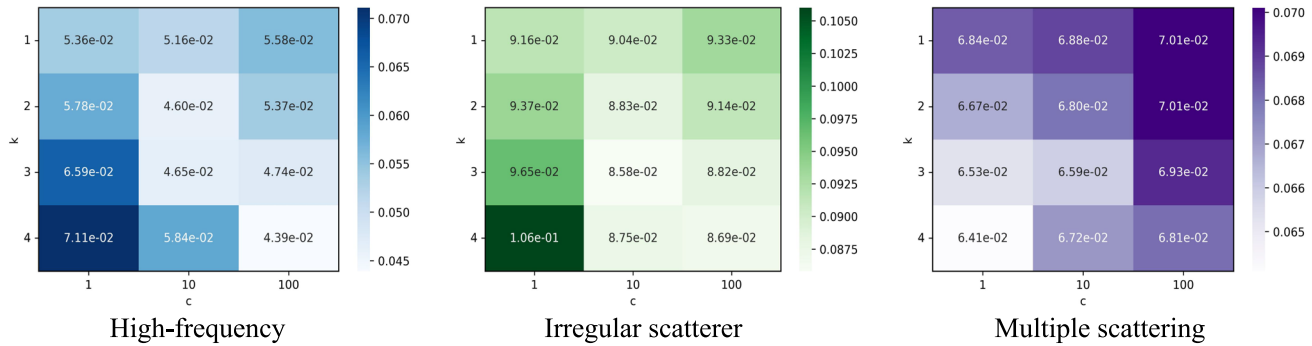
5.5 Sensitivity analysis

We investigated the impact of hyperparameters on PIKAN by varying network architectures, grid sizes, and RAD parameters.

When testing different network architectures with grid size 10, Table 7 shows the sensitivity of accuracy to network architectures in terms of high-frequency, irregular scatterer, and multiple scattering. When the network width increases from 5 to 10, the accuracy of all three scenarios improves significantly, indicating that a preliminary increase in width can enhance the model's expressive capacity. When the

Table 8 Prediction errors for different grid sizes on high-frequency, irregular scatterer, and multiple scattering problems

Grid Size	High-frequency			Irregular scatterer			Multiple scattering		
	Loss	rMAE	rMSE	Loss	rMAE	rMSE	Loss	rMAE	rMSE
5	0.0009	0.0619	0.0733	0.0074	0.1901	0.1831	2e-5	0.0722	0.0795
10	0.0003	0.0395	0.0563	0.0015	0.0925	0.0935	2e-5	0.0616	0.0704
15	8e-5	0.0362	0.0489	0.0012	0.0838	0.0866	6e-5	0.1119	0.1140


Fig. 9 Prediction errors for RAD method with different (k, c) on high-frequency, irregular scatterer, and multiple scattering problems

width increases to 15, the accuracy of irregular scatterer further improves, but that of high-frequency and multiple scattering decreases instead. This reflects differences in the requirements for the number of plane waves across different scenarios. Irregular scatterer require a wider network to cover more plane wave directions, while high-frequency and multiple scattering suffer from plane wave redundancy due to excessive network width, which in turn reduces accuracy. In addition, when the network depth increases, the complexity required to calculate plane waves in a single direction is better satisfied, thus maintaining low errors in all three scenarios. Experimental results demonstrate that $[2, 5 * [10], 2]$ performs better in all scenarios, hence we propose it.

When varying grid size with network architecture $[2, 5 * [10], 2]$, Table 8 shows the sensitivity of accuracy to grid size in terms of high-frequency, irregular scatterer, and multiple scattering problems. When the grid size increases from 5 to 10, the accuracy in all three scenarios shows improvement, indicating that increasing the grid size can enhance the model's ability to capture multi-scale features. When the grid size increases further to 15, the accuracy for high-frequency scattering and irregular scatterer improves further, while the accuracy for multiple scattering decreases. This reveals differences in the learned frequency components of plane waves across different scenarios. In the scenarios of high-frequency and irregular scatterer, a single plane wave covers more frequency components, thus requiring a larger grid for learning. However, due to the superposition principle, an excessively large grid is redundant for multiple scattering, leading to errors. We suggest a grid size of 10 for accuracy and computational efficiency in all scenarios.

Section 5.4 verified the effectiveness of the combination of the PIKAN model and the RAD method. To explain the sensitivity of RAD parameters (k, c) and offer guidelines for their selection, we use the RAD method with different parameters (k, c) to train PIKAN on three application scenarios, where $k = 1, 2, 3, 4$ and $c = 1, 10, 100$. In addition, the number of residual points $|T|$ and the number of sampling rounds R also influence the performance of the RAD method. In this experiment, we empirically set $|T| = 10000$, $R = 1$. Other parameters remain consistent with the previous section.

As shown in Fig. 9, for high-frequency scattering, the best rMSE of $4.39e-2$ is observed when $k = 4$ and $c = 100$; for irregular scattering, the best rMSE of $8.58e-2$ is observed when $k = 3$ and $c = 10$; for multiple scattering, the best rMSE of $6.41e-2$ was observed when $k = 4$ and $c = 1$. The experiments use three application scenarios and get extremely different optimal hyperparameters (k, c) . The results conclude that: to obtain an accurate result, one can use a large c with a large k for high-frequency scattering, and a small c with a large k for multiple scattering. For the irregular scatterer, priority should be given to using a large c with a large k .

This reveals the differences in the characteristics of the residual regions across different scenarios. Specifically, for high-frequency scattering, the wavelength is much smaller than the size of the scatterer, and the energy is mainly concentrated in the specular reflection direction and near the surface of the scatterer, while the scattered energy in other directions is relatively weak. Therefore, the coverage of its high-residual regions is small, and the ratio of residual values between these regions and other regions is relatively

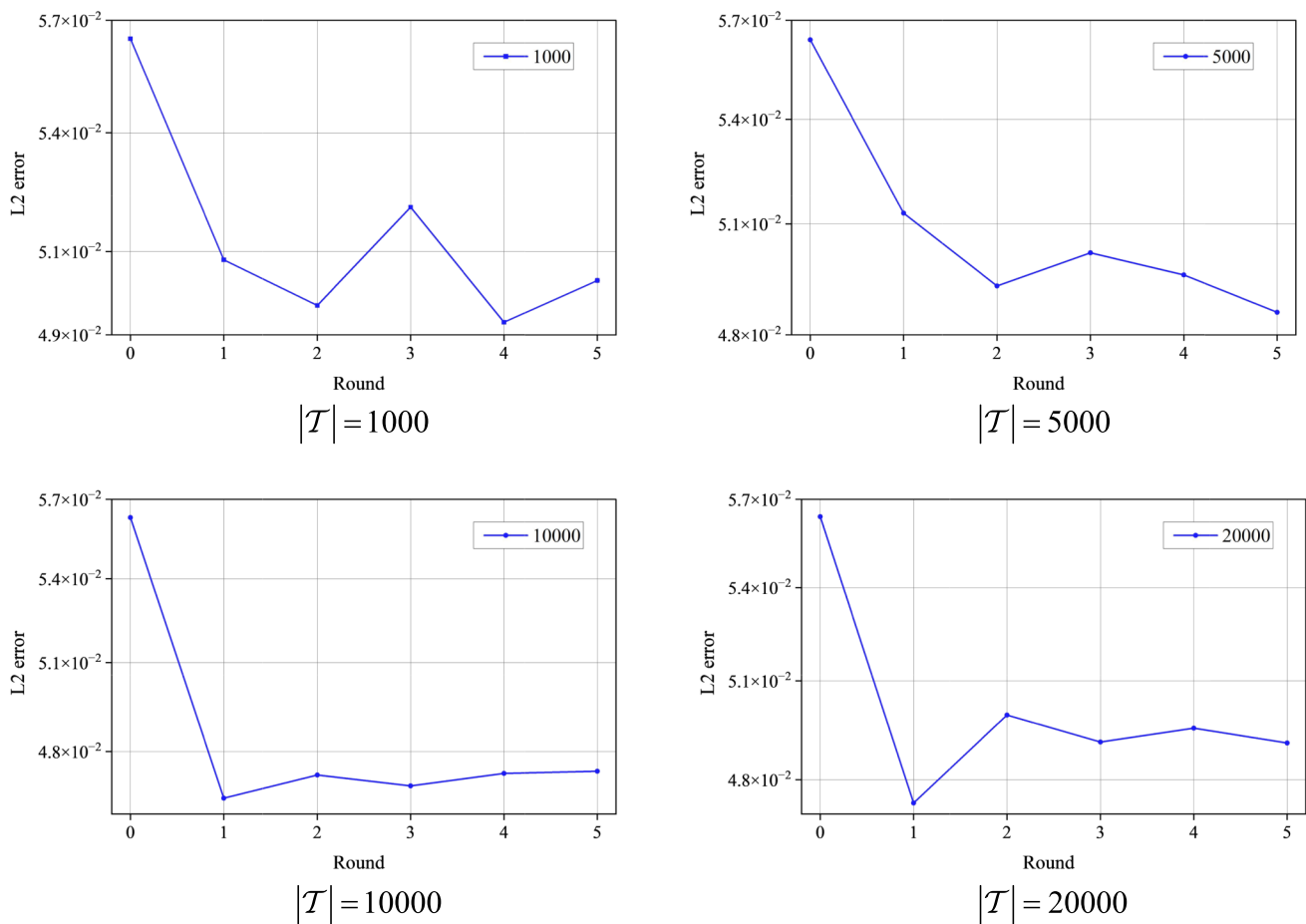


Fig. 10 Prediction errors for RAD method with different residual points $|\mathcal{T}|$ with increasing rounds R

large. For multiple scattering between multiple scatterers, residuals diffuse to the entire space where multiple scatterers are distributed. Meanwhile, the scattered energy becomes uniform due to mutual superposition and dissipation, with no obvious high-energy concentration areas. Therefore, the coverage of its high-residual regions is relatively large, and the ratio of residual values between these regions and other regions is relatively small. For low-frequency irregular scatterer, compared with high-frequency scattering and multiple scattering, both the coverage of high-residual regions and the ratio of residual values between these regions and other regions fall in the middle. Although the results showed that the RAD parameters (k, c) is sensitive to the scattering scenarios, the rMSE is accurate enough for all three scenarios in the case of RAD with $k = 3$ and $c = 10$.

Furthermore, we also analyzed the sensitivity of the number of residual points $|\mathcal{T}|$ and the number of sampling rounds R to prediction performance. The results are illustrated in Fig. 10. The experimental results show that in the initial phase, the error decay is steeper, which is because the RAD method accelerates the convergence in high-residual

regions. As the residuals decrease, the error decay rate slows down, which reflects the improved prediction accuracy of the PIKAN model in these regions. Additionally, as the round R increases, the ℓ_2 error fluctuates to varying degrees. This is because the characteristics of the residual region change, while the RAD parameters (k, c) remain at their initial settings, rather than adjust accordingly. Taking stability and accuracy into consideration, we select the optimal parameters as $|\mathcal{T}| = 10000$ and $R = 1$.

6 Conclusion

This study presents PIKAN, a new framework that combines KAN with Sine activation to overcome restrictions in local feature extraction with scattering field. By implicitly incorporating the plane wave expansion, PIKAN achieves great accuracy with fewer parameters and lower training costs across three application scenarios: the high-frequency case, irregular scatterer, and multiple scattering. Moreover, the Combination with the RAD method further enhances its performance.

When using the present method to solve higher-frequency scattering problems or those characterized by sharp edges, it may lead to issues such as loss function oscillation or gradient vanishing (or exploding), making the optimization process challenging or even nonconvergent. Meanwhile, as the number of scatterers increases in multiple scattering scenarios, the selected grid size and network architecture may prove insufficient. Note that most PINN models face similar limitations to PIKAN when solving complex scattering scenarios.

In future work, we aim to integrate wavelet packet modeling technology to better capture acoustic field features across different frequency components and spatial scales. Additionally, we are also interested in applying PIKAN to more complicated scattering scenarios, such as other typical scatterers (e.g., elastic scatterers) or complex environments (e.g., non-uniform media), which pose significant challenges due to their increased computational complexity and intricate acoustic wave interactions.

Data Availability No data was used for the research described in the article.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Tian C, Cheng T, Peng Z, Zuo W, Tian Y, Zhang Q, Wang F, Zhang D (2025) A survey on deep learning fundamentals. *Artif Intell Rev* 58:381. <https://doi.org/10.1007/s10462-025-11368-7>
- Chen X, Wei Z, Maokun L, Rocca P et al (2020) A review of deep learning approaches for inverse scattering problems (invited review). *Electromagnetic Waves* 167:67–81. <https://doi.org/10.2528/PIER20030705>
- Shen W, Zhang G, Yin H, Cong Y, Gu S, Yao B (2025) A mesh based improved plane wave expansion method for complex 2-d phononic crystals. *J Sound Vib* 606:119015. <https://doi.org/10.1016/j.jsv.2025.119015>
- Chen Y, Dal Negro L (2022) Physics-informed neural networks for imaging and parameter retrieval of photonic nanostructures from near-field data. *APL Photonics* 7(1):010802. <https://doi.org/10.1063/5.0072969>
- Nair S, Walsh TF, Pickrell G, Semperlotti F (2023) GRIDS-Net: Inverse shape design and identification of scatterers via geometric regularization and physics-embedded deep learning. *Comput Methods Appl Mech Eng* 414:116167. <https://doi.org/10.1016/j.cma.2023.116167>
- Sun Y, Tao H, Ni Y, Stojanovic V (2025) A generic single-source domain generalization framework for fault diagnosis via wavelet packet augmentation and pseudo-domain generation. *IEEE Internet Things J* 12(15):31629–31642. <https://doi.org/10.1109/JIOT.2025.3573752>
- Nair S, Walsh TF, Pickrell G, Semperlotti F (2025) Multiple scattering simulation via physics-informed neural networks. *Engineering with Computers* 41(1):31–50. <https://doi.org/10.1007/s00366-024-02038-3>
- Qu W, Gu Y, Zhao S, Wang F, Lin J (2024) Boundary integrated neural networks and code for acoustic radiation and scattering. *Int J Mech Syst Dyn* 4(2):131–141. <https://doi.org/10.1002/msd2.12109>
- Zhang R, Gao Y (2024) Learning scattering waves via coupling physics-informed neural networks and their convergence analysis. *J Comput Appl Math* 446:115874. <https://doi.org/10.1016/j.cam.2024.115874>
- Zhang D, Yuan Q, Meng L, Xia R, Liu W, Qin C (2026) Reinforcement learning for single-agent to multi-agent systems: from basic theory to industrial application progress, a survey. *Artif Intell Rev* 59:46. <https://doi.org/10.1007/s10462-025-11439-9>
- Qin C, Pang M, Wang Z, Hou S, Zhang D (2025) Observer based fault tolerant control design for saturated nonlinear systems with full state constraints via a novel event-triggered mechanism. *Engi Appl Artif Intell* 161 Part C, 112221. <https://doi.org/10.1016/j.engappai.2025.112221>
- Wu H, Ma Y, Zhou H, Weng H, Wang J, Long M (2025) Pro-PINN: Demystifying propagation failures in physics-informed neural networks. *arXiv preprint arXiv:2502.00803*, <https://doi.org/10.48550/arXiv.2502.00803>
- Zhang Z, Yan X, Liu P, Zhang K, Han R, Wang S (2023) A physics-informed convolutional neural network for the simulation and prediction of two-phase Darcy flows in heterogeneous porous media. *J Comput Phys* 477:111919. <https://doi.org/10.1016/j.jcp.2023.111919>
- Xu C, Liu D, Hu Y, Li J, Qin R, Zheng Q, Xiong J (2025) Sub-sequential physics-informed learning with state space model. *arXiv preprint arXiv:2502.00318*, <https://doi.org/10.48550/arXiv.2502.00318>
- Gao Y, Wang B, Lu J, Tian Z (2025) ML-PINN: A memory-efficient physics-informed Mamba-LSTM network for fast and accurate PDE solving. *Neurocomputing* 655:131446. <https://doi.org/10.1016/j.neucom.2025.131446>
- Peng J-Z, Hua Y, Aubry N, Chen Z-H, Mei M, Wu W-T (2024) Data and physics-driven modeling for fluid flow with a physics-informed graph convolutional neural network. *Ocean Eng* 301:117551. <https://doi.org/10.1016/j.oceaneng.2024.117551>
- Zhao Z, Ding X, Prakash BA (2023) PINNsFormer: A transformer-based framework for physics-informed neural networks. *arXiv preprint arXiv:2307.11833*, <https://doi.org/10.48550/arXiv.2307.11833>
- Liu Z, Wang Y, Vaidya S, Ruehle F, Halverson J, Soljačić M, Hou TY, Tegmark M (2024) KAN: Kolmogorov-arnold networks. *arXiv preprint arXiv:2404.19756*, <https://doi.org/10.48550/arXiv.2307.11833>
- Reinhardt E, Ramakrishnan D, Gleyzer S (2024) SineKAN: Kolmogorov-arnold networks using sinusoidal activation functions. *Frontiers in Artificial Intelligence* 7:1462952. <https://doi.org/10.3389/frai.2024.1462952>
- Raissi M, Perdikaris P, Karniadakis GE (2019) Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J Comput Phys* 378:686–707. <https://doi.org/10.1016/j.jcp.2018.10.045>
- Luo K, Zhao J, Wang Y, Li J, Wen J, Liang J, Soekmadji H, Liao S (2025) Physics-informed neural networks for PDE problems: A comprehensive review. *Artif Intell Rev* 58(10):1–43. <https://doi.org/10.1007/s10462-025-11322-7>
- Sun Z, Zhou Y, Tang S, Luo J, Zhao B (2024) Noise suppression zeroing neural network for online solving the time-varying inverse kinematics problem of four-wheel mobile manipulators with external disturbances. *Artif Intell Rev* 57(8):211. <https://doi.org/10.1007/s10462-024-10804-4>
- Qin C, Jiang K, Wang Y, Zhu T, Wu Y, Zhang D (2025) Event-triggered h_∞ control for unknown constrained nonlinear systems with application to robot arm. *Appl Math Model* 144:116089. <https://doi.org/10.1016/j.apm.2025.116089>

24. Zhang D, Wang Y, Meng L, Yan J, Qin C (2024) Adaptive critic design for safety-optimal FTC of unknown nonlinear systems with asymmetric constrained-input. *ISA Trans* 155:309–318. <https://doi.org/10.1016/j.isatra.2024.09.018>
25. Zhao B, Zhang S, Liu D (2024) Self-triggered approximate optimal neuro-control for nonlinear systems through adaptive dynamic programming. *IEEE Trans Neural Netw Learn Syst* 36(3):4713–4723. <https://doi.org/10.1109/TNNLS.2024.3362800>
26. Qin C, Hou S, Pang M, Wang Z, Zhang D (2025) Reinforcement learning-based secure tracking control for nonlinear interconnected systems: An event-triggered solution approach. *Eng Appl Artif Intell* 161 Part C, 112243. <https://doi.org/10.1016/j.engappai.2025.112243>
27. Wang S, Yu X, Perdikaris P (2022) When and why PINNs fail to train: A neural tangent kernel perspective. *J Comput Phys* 449:110768. <https://doi.org/10.1016/j.jcp.2021.110768>
28. Liu DC, Nocedal J (1989) On the limited memory BFGS method for large scale optimization. *Math Program* 45(1):503–528. <https://doi.org/10.1007/BF01589116>
29. Krishnapriyan A, Gholami A, Zhe S, Kirby R, Mahoney MW (2021) Characterizing possible failure modes in physics-informed neural networks. *Advances in Neural Information Processing Systems (NeurIPS)* 34:26548–26560
30. Sun Y, Tao H, Stojanovic V (2024) Autoregressive data generation method based on wavelet packet transform and cascaded stochastic quantization for bearing fault diagnosis under unbalanced samples. *Eng Appl Artif Intell* 138:109402. <https://doi.org/10.1016/j.engappai.2024.109402>
31. Braun J, Griebel M (2009) On a constructive proof of kolmogorov's superposition theorem. *Constr Approx* 30(3):653–675. <https://doi.org/10.1007/s00365-009-9054-2>
32. Shukla K, Toscano JD, Wang Z, Zou Z, Karniadakis GE (2024) A comprehensive and FAIR comparison between MLP and KAN representations for differential equations and operator networks. *Comput Methods Appl Mech Eng* 431:117290. <https://doi.org/10.1016/j.cma.2024.117290>
33. Somvanshi S, Javed SA, Islam MM, Pandit D, Das S (2025) A survey on Kolmogorov-Arnold network. *ACM Comput Surv* 58(2):1–35. <https://doi.org/10.1145/3743128>
34. Rigas S, Papachristou M, Papadopoulos T, Anagnostopoulos F, Alexandridis G (2024) Adaptive training of grid-dependent physics-informed Kolmogorov-Arnold networks. *IEEE Access* 12:176982–176998. <https://doi.org/10.1109/ACCESS.2024.3504962>
35. Nair S, Walsh TF, Pickrell G, Semperlotti F (2024) Physics and geometry informed neural operator network with application to acoustic scattering. *arXiv preprint arXiv:2406.03407*, <https://doi.org/10.48550/arXiv.2406.03407>
36. Luo K, Liao S, Guan Z, Liu B (2025) An enhanced hybrid adaptive physics-informed neural network for forward and inverse PDE problems. *Appl Intell* 55(4):255. <https://doi.org/10.1007/s10489-024-06195-2>
37. Wu C, Zhu M, Tan Q, Kartha Y, Lu L (2023) A comprehensive study of non-adaptive and residual-based adaptive sampling for physics-informed neural networks. *Comput Methods Appl Mech Eng* 403:115671. <https://doi.org/10.1016/j.cma.2022.11571>
38. Xu J, Jin B, Zhang J (2025) An eigen plane wave expansion method for reconstructing enclosed sound fields at low frequencies. *Appl Acoust* 231:110550. <https://doi.org/10.1016/j.apacoust.2025.110550>
39. Bozorgasl Z, Chen H (2024) Wav-KAN: Wavelet Kolmogorov-Arnold networks. *arXiv preprint arXiv:2405.12832*, <https://doi.org/10.48550/arXiv.2405.12832>
40. Aghaei AA (2024) rKAN: Rational Kolmogorov-Arnold networks. *arXiv preprint arXiv:2406.14495*, <https://doi.org/10.48550/arXiv.2406.14495>
41. Yu T, Qiu J, Yang J, Oseledets I (2024) SincKAN: Sinc Kolmogorov-Arnold network and its applications on physics-informed neural networks. *arXiv preprint arXiv:2410.04096*, <https://doi.org/10.48550/arXiv.2410.04096>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.